

Generation Interconnection System Impact Study Report

For Queue Project Q20-1554

Anza 230 kV

May 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1554. The Interconnection Customer proposes a 153.2 MW battery energy storage facility interconnecting at the Anza 230 kV substation in the Illinois area. The request was received on August 21, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,619,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

N Bonneille / Lake Charles Plant 100.0 kV: 594.3% loading (114 MVA) following the outage of the Ashlan Ave / Carlyss 230.0 kV line; DFAX = 7.1%.

For the loss of Carlyss / Ashlan Ave 230.0 kV, the Lake Charles Plant / Clintmoore 100.0 kV loads to 336.7% of its 179 MVA rating. Project DFAX: 7.1%.

Anza 230 kV/100 kV Transformer: 245.5% loading (135 MVA) following the outage of the Shoreline / Front Range 100.0 kV; DFAX = 13.4%.

For the loss of Anza / Ashlan Ave 230.0 kV, the Decatur Route 51 Substation / Cedar 100.0 kV loads to 156.5% of its 136 MVA rating. Project DFAX: 6.7%.

For the loss of Anza / Ashlan Ave 230.0 kV, the Asbury / Decatur Route 51 100.0 kV line loads to 151.2% of its 134 MVA rating. Project DFAX: 6.7%.

The Asbury / Brandy Branch 100.0 kV is overloaded to 146.7% of its 131 MVA emergency rating for the loss of the Anza / Ashlan Ave 230.0 kV Line. The project contributes 6.7% DFAX (10.2 MW) to this overload.

The Flambeau Hydroelectric Sta / Davis Crk 100.0 kV is overloaded to 146.6% of its 103 MVA emergency rating for the loss of the Anza / Ashlan Ave 230.0 kV. The project contributes 5.8% DFAX (8.9 MW) to this overload.

Leota / Brandy Branch 100.0 kV: 141.4% loading (128 MVA) following the outage of the Ashla Ave / Anza 230.0 kV; DFAX = 6.7%.

Anza / Sky Vly 100.0 kV line: 139.8% loading (142 MVA) following the outage of the Shoreline / Front Range 100.0 kV; DFAX = 6.3%.

Talentino Sw Sta / Amrad 100.0 kV: 139.1% loading (130 MVA) following the outage of the Anza / Ashlan Ave 230.0 kV; DFAX = 22.8%.

Flambeau Hydroelectric Sta 100-1 kV TX Unit 1: 139.0% loading (240 MVA) following the outage of the Anza / Ashlan Ave Substation 230.0 kV; DFAX = 19.0%.

The Moore Rd / Amrad 100.0 kV is overloaded to 136.5% of its 128 MVA emergency rating for the loss of the Anza / Ashlan Ave 230.0 kV. The project contributes 22.8% DFAX (35.0 MW) to this overload.

Cedar / Flambeau Hydroelectric Station 100.0 kV: 136.4% loading (170 MVA) following the outage of the Anza / Ashlan Ave 230.0 kV; DFAX = 6.7%.

The Shoreline / Flambeau Hydroelectric Sta 100.0 kV L is overloaded to 134.0% of its 520 MVA emergency rating for the loss of the Ashlan Ave / Anza 230.0 kV. The project contributes 31.6% DFAX (48.3 MW) to this overload.

The Dededo / Moore Rd 100.0 kV Line is overloaded to 133.3% of its 124 MVA emergency rating for the loss of the Ashlan Ave / Anza 230.0 kV. The project contributes 22.8% DFAX (35.0 MW) to this overload.

Shoreline / Front Range 100.0 kV: 132.8% loading (539 MVA) following the outage of the Ashlan Ave / Anza Substation 230.0 kV Line; DFAX = 31.6%.

For the loss of Anza / Ashlan Ave 230.0 kV, the Front Range / Socrates 19 100.0 kV loads to 132.8% of its 539 MVA rating. Project DFAX: 31.6%.

For the loss of Anza / Ashlan Ave 230.0 kV, the Dededo / Dairymans 100.0 kV loads to 130.9% of its 119 MVA rating. Project DFAX: 22.8%.

Oxbow / Blythe 100.0 kV line: 124.7% loading (673 MVA) following the outage of the Anza / Ashlan Ave 230.0 kV; DFAX = 7.1%.

Oxbow / N Onneville 100.0 kV: 124.3% loading (686 MVA) following the outage of the Anza / Ashlan Ave 230.0 kV; DFAX = 7.1%.

The Flambeau Hydroelectric Sta 230-1 kV XFMR is overloaded to 108.6% of its 290 MVA emergency rating for the loss of the Anza / Ashlan Ave Substation 230.0 kV. The project contributes 19.0% DFAX (29.1 MW) to this overload.

The Anza / Guardian 100.0 kV is overloaded to 108.2% of its 123 MVA emergency rating for the loss of the Shoreline / Front Range 100.0 kV. The project contributes 7.1% DFAX (11.0 MW) to this overload.

Carlyss / Boca Raton 230.0 kV L: 107.6% loading (724 MVA) following the outage of the Carlyss / Top Crop Substation 500.0 kV; DFAX = 65.3%.

Socrates 19 / Bear Canyon 100.0 kV: 100.9% loading (667 MVA) following the outage of the Carlyss / Ashlan Ave 230.0 kV; DFAX = 7.2%.

Bear Canyon / Clintmoore 100.0 kV: 100.9% loading (686 MVA) following the outage of the Carlyss / Ashlan Ave 230.0 kV L; DFAX = 7.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Bonneille / Lake Charles Plant 100.0 kV	Rebuild line with higher-capacity conductor	\$3,331,000	22% (\$720,000)
Lake Charles Plant / Clintmoore 100.0 kV	Rebuild line with higher-capacity conductor	\$10,602,000	22% (\$2,292,000)
Anza 230 kV/100 kV Transformer	Replace transformer with higher-rated unit	\$7,741,000	15% (\$1,178,000)
Decatur Route 51 Substation / Cedar 100.0 kV	Rebuild line with higher-capacity conductor	\$1,494,000	23% (\$350,000)
Asbury / Decatur Route 51 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,309,000	23% (\$775,000)
Asbury / Brandy Branch 100.0 kV	Rebuild line with higher-capacity conductor	\$3,413,000	23% (\$800,000)
Flambeau Hydroelectric Sta / Davis Crk 100.0 kV	Rebuild line with higher-capacity conductor	\$1,382,000	36% (\$494,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Leota / Brandy Branch 100.0 kV L	Rebuild line with higher-capacity conductor	\$5,268,000	23% (\$1,234,000)
Anza / Sky Valley 100.0 kV	Rebuild line with higher-capacity conductor	\$6,137,000	10% (\$614,000)
Tlentino Sw Sta Substation / Amrad 100.0 kV	Rebuild line with higher-capacity conductor	\$9,210,000	34% (\$3,121,000)
Flambeau Hydroelectric Sta 100-1 kV TX Unit 1	Replace transformer with higher-rated unit	\$2,367,000	20% (\$481,000)
Moore Rd / Amrad 100.0 kV	Rebuild line with higher-capacity conductor	\$7,010,000	34% (\$2,375,000)
Cedar / Flambeau Hydroelectric Station 100.0 kV	Rebuild line with higher-capacity conductor	\$446,000	23% (\$104,000)
Shoreline / Flambeau Hydroelectric Sta 100.0 kV L	Rebuild line with higher-capacity conductor	\$496,000	25% (\$126,000)
Moore Rd / Dededo Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$3,624,000	34% (\$1,228,000)
Shoreline / Front Range 100.0 kV L	Rebuild line with higher-capacity conductor	\$793,000	25% (\$202,000)
Front Range / Socrates 19 100.0 kV	Rebuild line with higher-capacity conductor	\$1,291,000	25% (\$329,000)
Dedeo / Dairymans 100.0 kV	Rebuild line with higher-capacity conductor	\$16,950,000	34% (\$5,743,000)
Oxbow / Blythe 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,741,000	22% (\$376,000)
North Bonneville / Oxbow Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$8,934,000	22% (\$1,929,000)
Flambeau Hydroelectric Sta 230 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$12,287,000	20% (\$2,497,000)
Guardian / Anza 100.0 kV	Reconductor line	\$401,000	28% (\$113,000)
Carlyss / Boca Raton 230.0 kV line	Reconductor line	\$13,097,000	54% (\$7,071,000)
Bear Canyon / Socrates 19 100.0 kV	Reconductor line	\$1,097,000	16% (\$176,000)
Bear Canyon / Clintmoore 100.0 kV	Reconductor line	\$1,809,000	16% (\$291,000)
	Total allocated to project		\$34,619,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.